

Multiplying & Dividing Fractions — Error Detective

Directions:

- Show every step of your work in the space under each problem.
- Write each answer as a fraction in simplest form (a mixed number is fine).
- Two problems show another student's work with a mistake in it. Find the mistake, then fix it.

1. Multiply: $\frac{2}{3} \times \frac{4}{5}$

Answer: _____

2. Multiply, and write your answer in simplest form: $\frac{5}{6} \times \frac{3}{10}$

Answer: _____

3. Divide: $\frac{3}{4} \div \frac{2}{5}$

Answer: _____

4. Jordan divides $\frac{5}{8} \div \frac{1}{2}$ like this:

$$\frac{5}{8} \div \frac{1}{2} \rightarrow \frac{8}{5} \times \frac{1}{2} = \frac{8}{10}$$

(a) Name the error in Jordan's work.

Answer: _____

(b) Find the correct quotient.

Answer: _____

5. A recipe problem: Mr. Diaz has 6 cups of flour. Each small loaf of bread uses $\frac{3}{4}$ cup of flour. How many loaves can he make?

Answer: _____

- 6.** Ava multiplies $\frac{2}{9} \times \frac{3}{4}$ like this:

$$\frac{2}{9} \times \frac{3}{4} \rightarrow \frac{2 \times 4}{9 \times 3} = \frac{8}{27}$$

- (a) Name the error in Ava's work.

Answer: _____

- (b) Find the correct product in simplest form.

Answer: _____

- 7.** A ribbon is $\frac{5}{6}$ meter long. It is cut into pieces that are each $\frac{1}{12}$ meter long. How many pieces are there?

Answer: _____

8. **Challenge.** Divide, and write your answer as a whole number: $2\frac{1}{2} \div \frac{5}{8}$

Answer: _____